Q1. Which of the following is a correct definition of a class ?
a) An instance of an object
b) A blueprint or prototype for creating objects.
c) A method in java
d) A package in java

Class is a blueprint while object is an instance of class

Q2. What is an object in java?
a) A ref to a class
b) A runtime entity
c) A method
d) An attribute

It is an Entity.

Q3. Which keyword is used to create a new instance of a class ?
a) new
b) this
c) instance
d) object

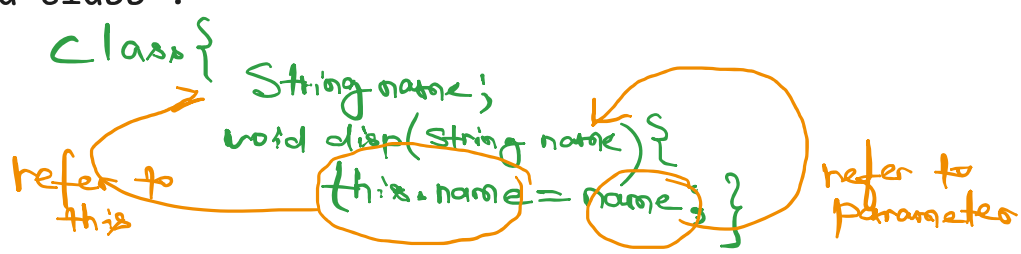
```
Child c = new Child();
```

Q4. What do we mean by the "state" of an object ?
a) Its behavior → Method defines behavior
b) The values assigned to the instance variable
c) Its method
d) Its interface

Q5. What are the main components of a class ?
a) Attributes and package
b) Objects and Reference
c) Methods and Attributes
d) Constructors and Destructors

Class consists of Methods like functions and attributes like fields, variables etc.

Q6. Which keyword is used to refer to the current instance of an object within a class ?
a) object
b) this
c) self
d) class



Q7. Which of these best describes "Encapsulation"?
a) Breaking a complex problem into simpler pieces
b) Wrapping up of data and methods into a single unit
c) inheriting properties from a parent class
d) implementing multiple interfaces.

definition

Q8. Which of these is Not an advantages of using classes and objects in java?

a) improved performance
b) Modular structure
c) Reusability of code
d) Easy to manage and control access

→ These are the advantages of class & Object

Q9. When a class inherits a property from another class, it is termed as ....?

a) Encapsulation
b) Polymorphism
c) Inheritance
d) Instantiation

→ Inheritance is the property in which a class Inherits the properties from another parent class.

Q10. Can a class have multiple constructors ?

a) yes
b) No

→ A class can have multiple constructors with same name and different parameters for Method Overloading.

Eg → class Sum {

Sum (int a, int b){
System.out.println("sum"+a+b);
}

Sum (int a, int b int c){
System.out.println("sum"+a+b+c);
}
}

both returns the sum